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Measuring attitude towards personal data for adaptive cybersecurity

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Abstract

Purpose – This paper presents an initial development of a personal data attitude (PDA) measurement instrument based on established psychometric principles. The aim of the research was to develop a reliable measurement scale for quantifying and comparing attitudes towards personal data that can be incorporated into cybersecurity behavioural research models. Such a scale has become necessary for understanding individuals' attitudes towards specific sets of data, as more technologies are being designed to harvest, collate, share and analyse personal data.

Design/methodology/approach – An initial set of 34 five-point Likert-style items were developed with eight subscales and administered to participants online. The data collected were subjected to exploratory and confirmatory factor analyses and MANOVA. The results are consistent with the multidimensionality of attitude theories and suggest that the adopted methodology for the study is appropriate for future research with a more representative sample.

Findings – Factor analysis of 247 responses identified six constructs of individuals' attitude towards personal data: protective behaviour, privacy concerns, cost-benefit, awareness, responsibility and security. This paper illustrates how the PDA scale can be a useful guide for information security research and design by briefly discussing the factor structure of the PDA and related results.

Originality/value – This study addresses a genuine gap in research by taking the first step towards establishing empirical evidence for dimensions underlying personal data attitudes. It also adds a significant benchmark to a growing body of literature on understanding and modelling computer users' security behaviours.

Keywords Privacy, Information disclosure, Information security modelling, Attitude scale, Cybersecurity, Personal data

Paper type Research paper

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1. Introduction

The amount of personal data being captured, collated, analysed and shared is growing every day. The trend is mediated by technologies such as smartphones, social networking and smart-meters (Manyika *et al.*, 2015). Previous work has highlighted that personal privacy is more vulnerable to erosions when contextual and personal information is gathered by pervasive computing systems (Beresford, 2005; Zhu *et al.*, 2009). Although personal data are recognized as a key issue requiring innovative cybersecurity measures within the digital economy, there are comparatively few studies exploring individuals' attitudes towards it. Iachello and Hong (2007) reviewed privacy-related literature within the context of human – computer interaction (HCI) and identified the need for a deeper understanding of individuals' attitudes towards the phenomena as a major challenge. Lederer *et al.* (2004) also acknowledged the fact that differing aspects of privacy pose a challenge for the design of usable systems. Essentially, there is not enough information available to guide stakeholders, including new technology designers and policymakers, in dealing with or addressing issues related to personal data. Innovative mobile information service providers are, for instance, faced with the question of how different users will respond to personal and context-aware services. Cybersecurity designers especially need to understand differing aspects of personal data issues to be able to develop systems that can adequately support values that constitute acceptable social behaviour.

The lack of empirically identified factors influencing individuals' digital security behaviour presents a major challenge to addressing the human component of cybersecurity. As technologies become more personalized and context-aware, there is a need to understand the interfaces and functionality required to accommodate individual differences in both use and attitude. The study presented in this paper is part of our ongoing work in information security behavioural research towards addressing the critical issue of leveraging knowledge about individual differences for the design of usable and adaptive cybersecurity. To identify relevant determinants of cybersecurity practices and predict individuals' security behaviour, we have presented an improved cybersecurity research model (Addae *et al.*, 2016) that integrates planned motivation theory (PMT) with TAM, and enables a wider variety of factors influencing cybersecurity behaviour to be explored (Figure 1). Our approach considers the personal data ecosystem (PDE) and how external factors such as users' prior experiences and demographic characteristics could shape an individual's beliefs regarding the benefits and consequences of certain security-related behaviour. We have identified the development of a personal data attitudes (PDAs) measurement scale as key to understanding and responding to people's views and attitude towards the data set around which digital technologies are built. Against this background, we determined to explore the different dimensions of attitudes towards personal data with the aim of developing a PDA scale, and verify the dimensions of attitudes toward different types of personal data. Thus, the paper focuses specifically on the development of a quantitative PDA measurement scale for the capture and analysis of attitudes across groups, contexts and data sets. Such an instrument could then be adopted in further pragmatic research activities to substantiate propositions in this area to offer theoretically informed guidelines for addressing personal data issues. Essentially, the results presented in this paper support the inclusion of personal data as a measurable variable in behavioural models being amplified for adaptive cybersecurity.

2. Personal data in the digital era

Technological advancement has the potential to enhance people's lives in many ways by facilitating the generation and sharing of knowledge. However, whenever we interact

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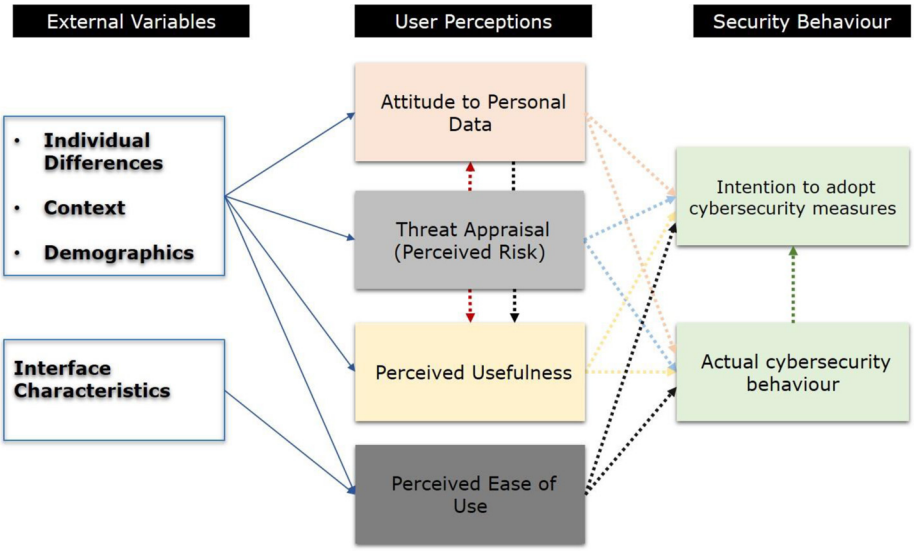

Figure 1. Proposed research model for the investigation and segmentation of cybersecurity behaviours

directly or indirectly with technologies, we leave behind data trails and digital footprints which can be used to generate information about our lives and activities. A considerable and increasing amount of information is gathered about us which are processed, stored, explored, shared, commercialized and potentially misused by both public and private individuals and/or organizations. This raises concerns around issues such as privacy, security and other digital asset rights (Joinson *et al.*, 2006). Consequently, when personal data are gathered, such as by pervasive computing devices, it is important to consider the range of potential implications for the individuals concerned. Previous research projects have explored personal data, mostly focusing on exploring people's attitude towards a single or limited subset of existing and/or near future technologies rather than the personal data itself. For instance, Brown (2013) adopted vignette-based survey to explore the social implications of data gathered through "Internet of Things" (IoT) in homes. They highlight a range of concerns about the technical systems but in some cases, it is unclear if these are due to the interface, data collection, display or data being collected. Although research approaches that explore users' attitudes to an identified technology are useful in assessing users' perceptions of a technology's usage of sensitive resources, it is also valuable to look at attitudes towards the data itself rather than the technologies through which it is created and accessed (Sharples *et al.*, 2013).

2.1 Underlying dimensions of personal data

A requisite preliminary stage in the creation of a validated measurement instrument is a consideration of the relevant construct dimensions. The relevant construct in this case, is individuals' attitudes towards personal data. In defining attitude to personal data, we adopt the general definition for attitude from the behavioural science literature and applied it to our context. Attitude to personal data here therefore refers to the behavioural tendency of an individual to negatively or positively evaluate the disclosure of a personal data. To successfully capture the factors influencing individuals' attitude to personal data, existing literature on the underlying dimensions of the concept was first reviewed. We started by

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considering available definitions of the object towards which attitude is being measured – personal data. In the UK Data Protection Act 1998 (Information Commissioner's Office, 2015), personal data are defined as:

> [. . .] data which relate to a living individual who can be identified (a) from those data, or (b) from those data and other information which is in the possession of, or is likely to come into the possession of, the data controller, and includes any expression of opinion about the individual and any indication of the intentions of the data controller or any other person in respect of the individual.

Personal information is also defined in the Privacy Protection Act of Australia as:

> [. . .] information or an opinion about an identified individual, or an individual who is reasonably identifiable: (a) whether the information or opinion is true or not; and (b) whether the information or opinion is recorded in a material form or not" [Australian Government, 2014, p. Subsection 6(1)].

The key component of these definitions is identifiability which means that certain level of personal information can be handled without legal implications as long as they are anonymized (Millard and Hon, 2012). Consequently, any information that can reasonably be directly or indirectly linked to an individual's identity qualifies as a personal identifiable information and requires careful handling (US Department of Labor, 2015). Information considered to be linkable to individuals includes medical, educational and financial records (McCallister, 2010). With this background, we decided to focus on measuring and comparing attitude towards four main types of personal data (social media, personal emails, financial and health records).

In the process of reviewing existing literature on personal data, we realized that, although the concept has now become a hot topic in the cybersecurity and privacy research community, much of the focus has been on developing technical and legal countermeasures. Studies exploring the individual behavioural elements of the concept are quite limited; hence, information on the structure of personal data as a psychometric construct is very scanty. Majority of the literature exploring personal data attitude based their studies on health and/or medical records. Rindfleisch (1997) for instance proposed and explained three concepts underlying health care information protection concerns – security, confidentiality and privacy. Wellcome Trust (2013) also measured and classified peoples' attitude to health data into identity, attention and control concerns. Consequently, three major themes were initially identified in the literature as dimensions of public views to personal data. These include issues related to security, risk/benefit trade-off and privacy/confidentiality. These dimensions mostly overlap with the eight underlying principles of the fair information practices outlined in the USA: PPSC (1977) and the (Organisation for Economic Co-Operation and Development, 2002) reports. The attention concern described by Wellcome Trust (2013) where users express mixed blessings about giving away their personal information, falls within the object of purpose specification principle. Thus, a user may give out personal information for an immediate benefit but may not be sure of the future implications of such an action. The principle of use limitation extends to the case where a user may be concerned about their identity being abused once their personal data are disclosed to access a service. The principle of collection limitation aims to addresses security and privacy concerns, by providing a framework for limiting the amount of specific PII that can be collected within specific contexts. Most definitions of privacy refers to people's ability to control the terms under which their personal information are acquired and used (Westin, 1967; Rindfleisch, 1997; Earp and Payton, 2006). Collection limitation is about having a limit to the collection of personal data and doing so in a legalized manner. The dimension of

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control is illustrated as a personal data concern with users' comments on losing their freewill and not being able to go for "opt-outs" as illustrated in the Wellcome Trust report.

The World Economic Forum (WEF, 2013, p. 3) hosted a global dialogue on the emerging issues surrounding the collection and use of personal data by clustering the eight principles into three main themes: "protection and security, accountability and rights and responsibilities for using personal data". Security here has to do with the integrity, availability and controlled access to information. This clearly encompasses the control concern identified by Wellcome Trust. The concept of confidentiality described by Rindfleisch (1997) directly overlaps with the principle of use limitations as they both has to do with the release of information when accessing a service (in this case health care) in a legal manner that limits the extent to which they may further be used or released. Rindfleisch (1997) refers to privacy as the right and desire of a person to control the disclosure of personal health information. This description is very much synonymous with the general definitions of privacy identified in the literature as mentioned earlier. Though there are limited studies attempting to measure attitude to personal data directly, several studies have explored privacy concerns. Smith *et al.* (1996) developed the Concern for Information Privacy (CFIP) scale which identified and measured four factors (collection, errors, secondary use and unauthorized access to information) as the dimensions of a person's privacy concern about organizations. Malhotra *et al.* (2004) later identified three aspects of information privacy, namely, attitudes towards the collection of personal information, control over personal information and awareness of privacy practices. More recently, Hong and Thong (2013), consolidated existing conceptualization of Internet Privacy Concerns (IPC) including Smith *et al.* (1996) and Malhotra *et al.* (2004) in their study and came up with six first-order factors of IPC – collection, secondary usage, errors, improper access, control and awareness. When reviewing these IPC measurement instruments, this article focuses on the broader PDE by identifying constructs that goes beyond privacy concerns to encompass other construct domains. Table I summarizes the construct domains generated from the review of existing literature pertaining to attitudes toward personal data.

3. Scale development

This paper describes a multistage scale development study conducted to develop a measurement instrument for PDA. Fundamentally, the issues of reliability and validity underpin the development of an attitude scale right from item generation, the theoretical deduction of a factor structure and resulting psychometric analysis. Previous scale development studies and recommendations from Hinkin (1995) on how to improve the scale development process provided guidance for the research. Subsequently, an iterative process (Figure 2) is adopted to assess the consistency of the scale items with the dimensions of PDAs identified in the literature. This section describes the procedures involved in developing and assessing the initial version of the PDA instrument.

3.1 Generation of initial Pool of PDA items

Following an examination of existing personal data related literature, definitions and surveys (including Joinson *et al.*, 2006; Lang *et al.*, 2009), 50 Likert-type attitudinal items related to the 10 construct domains identified was generated. The items drawn from these sources were rephrased to mainly reflect attitudes towards four different types of personal data, namely, email, social media (online personal data); financial and health (offline personal records). Because we did not find a measurement scale specifically designed for attitude towards personal data in the existing literature, an exploratory study was

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Construct domain	Construct definition	Relevant literature
Awareness	Conscious of issues surrounding the disclosure and use of personal data. Being aware of the general interest in personal data	Chen and Ismail (2013), Wang and Yu (2015), Wellcome Trust (2013), Singleton *et al.* (2008), Robling *et al.* (2004), Malhotra *et al.* (2004)
Collection	Concerns over the amount of specific types of personal data collected/disclosed within a specific context	Malhotra *et al.* (2004), Smith *et al.* (1996), Sheng *et al.* (2008)
Responsibility	The perception of who should be held accountable for the protection of personal data (individuals vs data protection authorities)	Hon *et al.* (2012), World Economic Forum (2013)
Cost-Benefit	Perceived risks and benefits of linking personal information disclosed to different kinds of data for different purposes. Thus, the positive or negative effects of sharing personal data in general	Dinev and Hart (2006), World Economic Forum (2013), Wellcome Trust (2013)
Privacy/ Confidentiality	Perception of control concerning who can have access to one's personal information and the need for anonymity	Malhotra *et al.* (2004), Dinev and Hart (2006)
Protection/ Security	Perception of the relevance and adequacy of existing technical and legal protective measures to ensure integrity, confidentiality and reliability of personal data	World Economic Forum (2013), Belanger *et al.* (2002), Lang *et al.* (2009), Dinev and Hart (2006)
Exposure/ Experience	The impact of past experience with disclosure of personal data	Hui *et al.* (2007), Awad and Krishnan (2006)
Protective behaviour	Actual steps taken to demonstrate a positive attitude to personal data such as choosing to opt-out, adoption of cybersecurity measures etc	Pattinson *et al.* (2016), Buchanan *et al.* (2007)
Interest	The level of attention or complacency exhibited towards personal data protection	Wang and Yu (2015), OPM (2015)
Sensitivity	the perception that a specific set of personal data will have damaging effect if exposed. The degree of risk associated to the disclosure of a personal data	Pearson (2013), Bansal and Gefen (2010)

Table I. Potential constructs of attitude to personal data

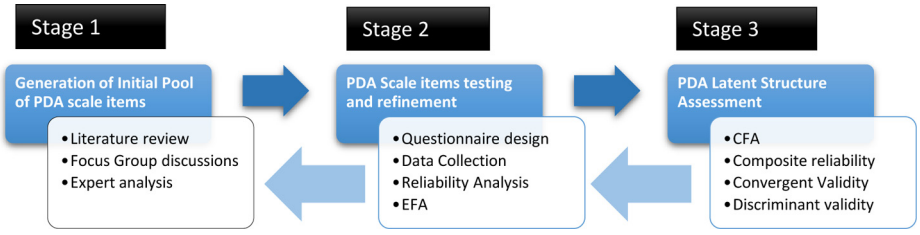

Figure 2. Iterative workflow adopted for the development of a PDA measurement scale

conducted as a preliminary step towards generating the PDA scale items. To ensure the content validity of construct domains predetermined from the literature review, focus group discussions were held using open-ended questions to elicit themes that constituted individuals' view on the four types of personal data. In total, 50 additional items were generated based on responses emerging from the focus group discussion on personal data matters. This resulted in an initial pool of 100 items serving as the basis for the PDA measurement. Further, 64 items were eventually dropped following an exercise to merge

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similar themes and convert the 100 items into generic personal data statements. Thus, those that were obviously pointing to a specific type of personal data (e.g. I feel my profile information on any social media is much secured) were discarded.

3.2 Scale specification and refinement

The research team reviewed the remaining 46 items based on the 10 construct definitions in Table I. We adapted items from previous privacy-related measurement scales to fit the PDA context as much as it was possible to do so. Items generated for the experience and sensitivity construct domains were dropped as they were mainly measured with categorical rather than scale data in previous studies. To minimize the tendency for respondents to provide the same responses or agree with statement due to acquiescent response bias, some items were worded negatively. To do this the direction of each statement (positive or negative attitude to personal data) needed to be determined. Items that were categorized as unable to judge statements were discarded. For instance, we could not indicate whether a statement like "I do not mind sharing such information with family and friends" is a positive or negative attitude towards personal data. Consequently, the items were further reduced to 34 PDA statements.

Each of the items was a statement to which people were asked about their level of agreement on a five-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). Thus, attitude to personal data was quantified as a continuous variable. Questions relating to some of the theoretically distinct aspects of personal data discussed earlier were included. For instance, items relating to security/identity (e.g. "There should be stronger laws to protect such personal data"), control/privacy (e.g. "I consider the privacy policy of institutions where I give out such personal details") and possible benefits of surrendering personal data (e.g. "I am happy to provide such personal details to support government policy and decision making"). The scale involved both positively and negatively worded items. To ensure that a higher numbered response on the Likert scale would represent positive attitudes, all negatively worded items were reversed before the data were analysed.

3.3 Data collection

A Web-based questionnaire was developed based on the 34 PDA items. The questionnaire had two main sections of demographics and the attitude to personal data items with four research design conditions (personal email, social media, financial and health data). Essentially, participants were randomly presented the generic statements with respect to one of the four types of personal data outlined above until they were almost evenly distributed across the groups. A non-probability sampling approach was adopted to collect responses from participants over the internet. Email invitations and anonymous link to the survey was posted on social media sites (Facebook, Twitter and LinkedIn). To further expand the sample size, snow-balling and convenience sampling techniques were also used. After several follow-up rounds, in total, 256 responses were received out of which 247 completed data sets was extracted at the data cleaning and preparation stage. Of the 247 respondents, 127 (51.4 per cent) were male and 120 (48.6 per cent) were female. The average age of the sample was 36 years (range: 17-67 years). As presented in Table II, most respondents use social media (90.7 per cent) out of which most of them preferred to use Facebook (47.8 per cent). Respondents who have not had previous experience with personal information misuse formed a significant portion of the sample population (83 per cent).

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Sample characteristic	*n*	Distribution (%)	Sample characteristic	*n*	Distribution (%)
Gender			Favourite social media		
Male	127	51.4	Facebook	118	47.8
Female	120	48.6	Twitter	36	14.6
Age**			None	23	9.3
17-24	47	19.0	WhatsApp	20	8.1
25-35	159	64.4	Instagram	15	6.1
36-67	41	16.6	LinkedIn	12	4.9
Average age (36)			Others*	23	9.3
Uses social media			Have email account		
Yes	224	90.7	Yes	245	99.2
No	23	9.3	No	2	0.8
Number of social media subscriptions per participants					
1-2	129	52.2	Had prior experience with personal data misuse		
3-4	74	30.0	Yes	64	25.9
≥5	44	17.8	No	183	74.1

Notes: *Includes ten different social media such as reddit, WeChat, YouTube, etc; ***given in years*

Table II. Participant demographics

4. Analysis and results

4.1 Reliability analysis

The 34 items of the attitude to personal data measure were subjected to an iterative scale purification procedure. To determine the internal consistency of the items, the most widely used reliability method of computing the Cronbach's alpha was adopted (Papanastasiou, 2005). This yielded 0.926, indicating a high reliability of the 34-item scale. A correlation matrix generated with SPSS was scanned to check the pattern of relationships among the items. There is no singularity in the data as all the correlation coefficients were less than 0.8 (Field 2013). Another commonly accepted procedure used to further assess the internal consistency of the items was the item-to-total correlations. A close look at the inter-item correlations and item-to-total correlations of each item revealed inadequate performance of some items. If the items are all measuring attitude to personal data, then each item ought to correlate with the total score from the questionnaire (Rattray and Jones, 2007). The correlation between participants' score on an item and the sum of their scores on all the items is represented by the r value. Three of the items that poorly correlated ($r < 0.4$) were removed from the PDA items. This conforms to the generally accepted rule-of-thumb that item-to-total correlations should exceed 0.30 (Van der Heijden and Sørensen, 2002). After deleting the items that fell below this standard, 31 items remained in the pre-final version of the questionnaire. The 31 items were then subjected to a separate reliability test. This resulted in an acceptable item-to-total correlation but the Cronbach alpha remained at 0.926.

4.2 Exploratory factor analysis

An exploratory factor analysis (EFA) can help to empirically determine how many constructs, or factors, underlie the set of PDA items (Child, 2006; Rattray and Jones, 2007). To determine the appropriateness of the factor analysis, a Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity were first examined. The KMO value for the data set was 0.899 and Bartlett's test of sphericity was significant ($p <$

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0.000), indicating that the factor analysis was appropriate. The initial EFA analysis produced a pattern matrix consisting of seven factors based on eigenvalues greater than 1 and accounted for 64.135 per cent of the total variance. Only one item loaded on the seventh factor and most of the items generated for protective behaviour and the interest scale were loading together. To minimize errors associated with under-extraction and/or over-extraction, a mixed approach based on both the eigenvalue and scree plot results was adopted. For instance, under-extraction error could lead to inconsistencies in the analysis and interpretation of the results (Reise *et al.*, 2000). In effect, based on the initial factor extraction and the results from the scree test (Figure 3), the factor analysis was repeated to extract a six-factor solution with an oblique rotation method to allow the obtained components to correlate (see Table III). This supports the assumption that PDA dimensions are related yet distinct from each other (Lee and Comrey, 1979). The final six-factor model emerging from the 31 PDA items explained 63.983 per cent of the total variance. Factor 1 contained the items measuring protective behaviour and interest, while Factor 2 focused on the items measuring privacy. Factor 3 involved the items measuring cost–benefit and Factor 4 contained the items measuring awareness. Factor 5 consist of the items measuring responsibility, and Factor 6 involve those items measuring security.

4.3 Assessment of the factor structure

Next, confirmatory factor analysis (CFA) was conducted using AMOS 23 to examine the factor structure obtained from the EFA. Maximum likelihood parameter estimates were used to examine the model fit. Researchers are required to report several fit indices in order

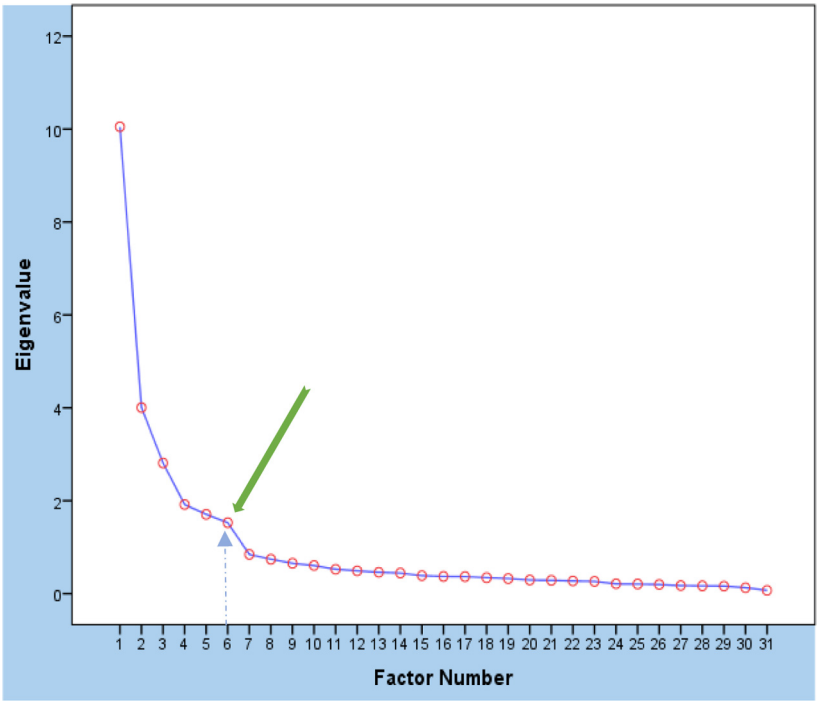

Figure 3.
Scree plot of factors underlying the PDA scale

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Scale items	Standardised factor loadings	SMC	Composite reliabilities	AVE
Protective Behaviour/Interest (BEH)			0.92	0.67
I always optimize my privacy settings when I create an online profile	0.90	0.81		
I consider the privacy policy of institutions where I give out such personal details	0.84	0.71		
I regularly look out for new policies on personal data protection	0.84	0.70		
I would opt out of a service due to privacy issues related to such data - *Recoded*	0.82	0.67		
I watch for ways to protect my personal information from unauthorized access	0.80	0.65		
I use security tools such as firewalls, encryption and other security settings to protect my private data	0.71	0.51		
Privacy/Confidentiality concerns (PRI)			0.91	0.63
I am concerned about my information online being linked to my publicly available offline data	0.88	0.78		
I'm concerned that too much personal information about me is being collected by so many organizations	0.82	0.67		
I worry about such information getting exposed – *Recoded*	0.81	0.66		
It usually bothers me when am asked to provide such personal information	0.80	0.65		
I am concerned about sharing such personal information because it could be used in a way I did not foresee	0.74	0.55		
It usually bothers me when I do not have control over decisions about how my personal information is collected, used, and shared	0.67	0.45		
Cost–Benefit (COB)			0.93	0.73
The risk posed to me if such personal information is exposed outweighs the benefits of sharing it	0.92	0.85		
In general, my need to obtain services is greater than my concern about privacy	0.87	0.76		
I am happy to provide such personal details to support government policy and decision making	0.84	0.70		
I value the personalized services I received from providing such personal data	0.84	0.71		
Such personal information can be used to victimize people - *Recoded*	0.80	0.64		
Awareness (AWA)			0.86	0.504
Such details about me are of value to external organizations	0.84	0.70		
Researchers need my consent to access my personal data – *Recoded*	0.78	0.61		
Service providers do not have the right to sell personal details of their users – *Recoded*	0.68	0.47		
Companies seeking information should disclose the way the data are collected, processed, and used	0.68	0.46		

(*continued*)

Table III. Scale items and CFA results

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Scale items	Standardised factor loadings	SMC	Composite reliabilities	AVE
It usually bothers me when commercial/government organizations seeking such information do not disclose the way the data will be processed, used and secured	0.66	0.44		
It is very important to me that I am aware and knowledgeable about how my personal information will be used	0.59	0.35		
Responsibility (RES)			0.84	0.57
Responsibilities lie with data handlers to ensure consistent and helpful applications of use (and not abuse)	0.78	0.61		
Designated data protection authorities are responsible for ensuring such personal data are only processed in accordance with the data protection regulations	0.77	0.59		
I would welcome the opportunity to pay for the privacy of such personal details – *Recoded*	0.75	0.57		
I would prefer to be personally responsible for the security of such personal data – *Recoded*	0.73	0.53		
Security (SEC)			0.85	0.59
I am concerned that databases that contain my personal information are not protected from unauthorized access	0.78	0.61		
I worry about wrong information being linked to my identity due to security breaches	0.77	0.59		
I worry about such information getting missing due to lack of adequate security measures	0.77	0.59		
I believe stronger security measures are required to ensure the correctness of such personal information	0.74	0.55		

Table III.

to characterize the fitness of a model correctly with the following boundary scores indicating good model fit – χ^2/df = 2.0-5.0, RMSEA < 0.06 [<0.05, < 0.08] and CFI > 0.94 (Hooper *et al.*, 2008). Results of the CFA (χ^2/df = 1.802, RMSEA = 0.057 and CFI = 0.940) indicate that the measurement model fits the data quite well. The six-factor solution is therefore supported by the CFA results. As observed in Table III, the reliability of the PDA scale is supported by the composite reliability estimates (>0.7) ranging from 0.84 to 0.93, indicating a good internal consistency of the multiple items for each construct in the model. The convergent validity was evaluated by checking all the values of average variance extracted (AVE) and the factor loadings. As shown in Table III, the estimated AVEs of all the PDA dimensions were all greater than the unexplained variances (>0.5) and all the factor loadings for the six constructs were above 0.5 and were significant for the individual items. The examination of the AVEs together with the factor loadings therefore confirmed the convergent validity of the PDA latent constructs. To investigate the discriminant validity of the PDA scale, the suggestions provided by Bertea and Zait (2011) regarding use of AVE analysis were followed. Accordingly, the value of the AVE for each construct should be at least 0.50 and the square root of each construct's AVE should be much larger than the correlation of the specific construct with any other constructs. The results from the AVE

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analysis presented in Table IV show that the shared variance between any two constructs was not greater than the square root of the corresponding AVEs. In summary, the assessments carried out to verify the measurement model yielded evidence for the reliability of the latent constructs.

4.4 Analysis of variance

The data were also analysed to determine whether there are differences between the four types of personal data. A series of multivariate analysis of variance (MANOVA) was performed using factor scores for participants' responses on the six constructs identified above as the dependent variables (DVs); and type of personal data (email, health data, financial records and social media), prior experience (Figure 4) and sensitivity responses (Yes and No) as the independent variables (IVs) (Figure 5). Bartlett's approach was used to compute the factor scores in SPSS to obtain unbiased estimates of the true factor scores (Everitt and Howell, 2005). As shown in Table V, the multivariate test using Wilks' lambda revealed an overall significant effect for data type as an IV on the DVs ($F = 1.77$) at $p = 0.025$ ($p < 0.05$). Univariate analysis for the effect of type of personal data in the survey, presented in Table VI, significantly predicted responses related to *behaviour* ($p < 0.05$) and *privacy concerns* ($p < 0.05$). Finally, post hoc testing (with least significant difference [LSD]) revealed highest *protective behaviour* scores for health data (mean = 24.87, SD = 3.03) and email (mean = 24.19, SD = 3.44), with health data being significantly higher (at $p < 0.05$) than

Constructs	BEH	PRI	COB	AWA	RES	SEC	*M*	*SD*
BEH	*0.82*						4.01	0.59
PRI	0.40*	*0.79*					3.51	0.79
COB	0.27*	0.18*	*0.86*				2.32	0.84
AWA	0.41*	0.53*	0.47*	*0.71*			2.94	0.71
RES	0.32*	0.19*	0.51*	0.34*	*0.76*		2.80	0.85
SEC	0.60*	0.53*	0.24*	0.50*	0.29*	*0.77*	3.95	0.58

Note: *Correlation is significant at the 0.01 lever (two-tailed)

Table IV. Factor inter-correlations

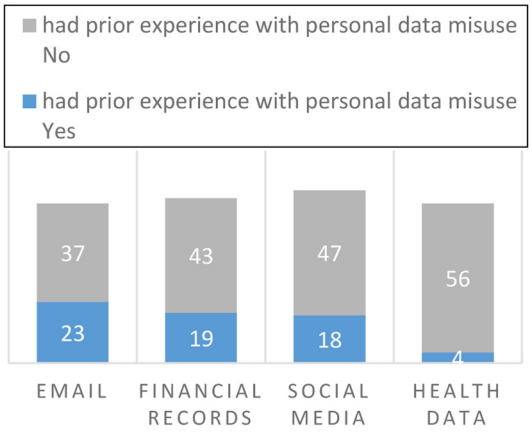

Figure 4. A frequency distribution of responses per personal data type and prior misuse experience

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social media (mean = 23.51, SD = 3.72). No significant mean difference in privacy concerns scores were obtained between the four types of personal data but the highest score was for *financial records* (mean = 21.58, SD = 5.16). The implications of these findings are discussed in the next section.

4.5 Cluster analysis

To identify homogeneous groups in the data set, we used a Two-Step clustering approach to cluster participants based on the six PDA factor scores computed. Two-Step clustering was chosen due to its ability to automatically determine the optimal number of clusters in the data set using an agglomerative hierarchal method (Mooi and Sarstedt, 2010). BIC (Schwarz's Bayesian Information Criterion) was used first to determine the number of

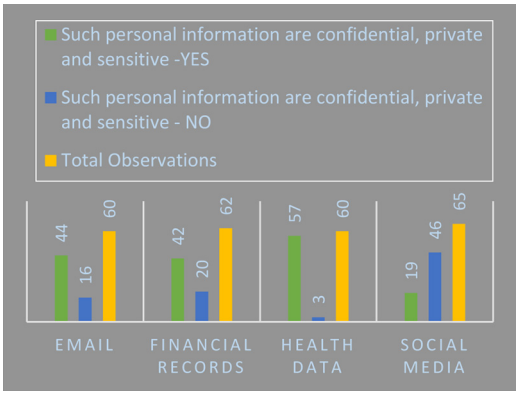

Figure 5.
A Frequency distribution of responses per personal data type and perception of sensitivity of data type

Table V.
Multivariate ANOVA done on the factor scores for the 6 PDA constructs across four types of personal data

	Value	*F*	Hypothesis df	Error df	Significance
Pillai's trace	0.132	1.759	18	690	0.026 $p < 0.05$
Wilks' lambda	*0.873*	*1.77**	*18*	*645.367*	*0.025* $p < 0.05$
Hotelling's trace	0.141	1.778	18	680	0.024 $p < 0.05$

Notes: Each *F* tests the multivariate effect of type of personal data. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means; *Wilk's lambda and the difference is significant

Table VI.
Univariate test done on each of the six DVs for the types of personal data

Dependent variable		Sum of squares	df	Mean square	*F*	Significance
Dimensions of PDA	*Behaviour*	*8.624*	*3*	*2.875*	*2.887*	*0.036*
	Privacy	*9.478*	*3*	*3.159*	*3.042*	*0.030*
	Cost_Benefit	0.420	3	0.140	0.133	0.940
	Responsibility	3.733	3	1.244	1.128	0.338
	Security	7.224	3	2.408	2.073	0.105

Notes: The *F* tests the effect of type of personal data. This test is based on the linearly independent pairwise comparisons among the estimated marginal means

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clusters, then AIC (Akaike's Information Criterion) was used. Figure 6 summarizes the results obtained with BIC, which do not differ from those obtained with AIC. The cluster centroids facilitated interpretation of the resultant cluster solution. The resulting clusters are labelled based on the protective behaviour factor which happens to be the overall important predictor variable (see Figure 7). The first cluster, which is the largest (57.9 per cent), contains participants with responses indicating a general privacy consciousness (mean privacy factor score = 0.11) and a somewhat protective attitude towards their personal data, with a mean score of 0.06 on the protective behaviour factor. The second cluster contain 23.9 per cent of the total participants with responses indicating

Cluster	**1**	**2**	**3**
Label	**Protective**	**Very Protective**	**Not Protective**
Description	Respondents' highest factor score was on Privacy but very low on Awareness, Cost_Benefit and Responsibilty	The respondents scored highly on all 6 factors especially on Awareness	Respondents in this cluster scored negatively on all six factors especially on Behaviour
Size	57.9% (143)	23.9% (59)	18.2% (45)
Inputs (PDA Factors and Mean Scores)	Cost-Benefit -0.15	Awareness 1.01	Awareness -1.17
	Responsibility -0.16	**Behaviour 0.84**	**Behaviour -1.30**
	Privacy 0.11	Responsibility 0.97	Cost-Benefit -0.85
	Security 0.08	Cost-Benefit 1.00	Privacy -1.13
	Behaviour 0.06	Security 0.66	Security -1.12
	Awareness -0.05	Privacy 0.58	Responsibility -0.77

Figure 6. Cluster distribution showing cluster centres sorted by within-cluster membership predictor importance

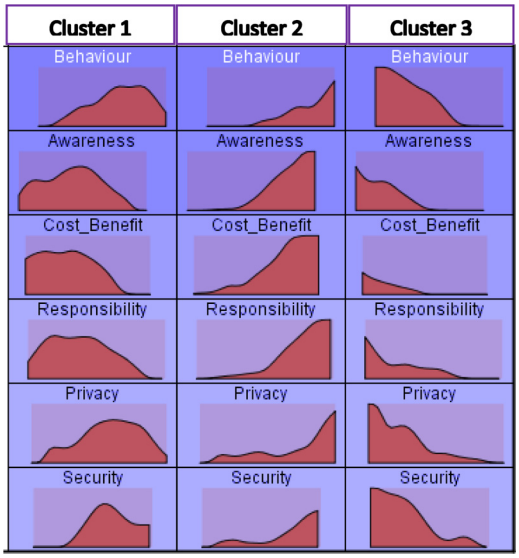

Figure 7. Comparison of the relative distribution of PDA factor scores sorted by overall cluster membership predictor importance for the three clusters

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high awareness on personal data issues, and obtained the highest mean score on the protective behaviour factor (0.84). The third cluster is the smallest (18.2 per cent) and consist mainly of participants whose responses indicate a general lack of interest in personal data-related issues, and scored the lowest on all six PDA factors, especially on protective behaviour (Mean = −1.30). The cluster centres presented in Figure 6 are sorted to highlight the within-cluster predictor (PDA factors) importance, while Figure 7 highlights the overall cluster membership predictor importance. A multivariate analysis conducted using the clusters as independent variable and the six factors as dependent variables shows significant differences in PDAs across the segments (Wilks' lambda = 0.196, $p > 0.000$). The results of a univariate F test revealed the clusters were significantly different on all PDA segment predictor factors. The discriminant analysis conducted based on the three clusters indicated that the model could correctly classify 94.3 per cent of respondents into groups.

5. Discussion

This study aimed to establish the dimensions required for the development of a reliable and valid PDA measurement scale. A PDA scale was successfully developed and verified based on the iterative scale development procedure adopted. Although the focus of this study was on attitude to personal data, the scale development process included an examination of both privacy and personal data literature, and the related regulations. The research results show that six constructs (protective behaviour/interest, privacy, cost-benefit, awareness, security and responsibility) are important components capable of differentiating individual attitude towards personal data. The reliability analysis yields support for the PDA scale's ability to reliably measure individual differences. The MANOVA results show that individuals' attitudes may vary based on the type of personal data. The results also suggest that participants who provided responses to the scale items based on health records generally viewed this data set as sensitive, and tended to score higher on the protective behaviour construct. Conversely, those who provided responses in relation to their personal social media profile data mostly did not view it as sensitive, and tended to score the lowest on both protective behaviour and privacy concerns. Interestingly, there were many social media participants who had had prior experience with personal data misuse (18) as compared to health data participants (4) (Figure 4).

The study also examined the relationship between the six extracted factors. Overall, the strongest relationship existed between the construct of privacy concerns and awareness as well as privacy and security. Thus, privacy correlates positively and reliably with both attitudes relating to awareness and security concerns. A possible interpretation here is that people who are more concerned about their privacy being breached tend to be more aware of the potential value of their personal data, and the risk factors associated with it. In general, all the constructs correlate significantly with each other. Notwithstanding the assumption that each construct measured completely different aspects of individuals' personal data attitude, the level of correlation between them is an indication of the conflicting conceptions people generally have on the subject. The results from the cluster analysis show that, despite the diversity in attitudes towards personal data, there is a relatively small number of compatible groups of users sharing similar attitudes and behaviours. Research reported by Norberg *et al.* (2007) and (Sato, 2010) indicates that even though consumers express concerns for their personal data, they generally do not take enough protection measures towards it. Acquisti and Gross (2006) also compared stated attitudes with actual behaviours of members of online social networks (OSNs) and found that reported privacy attitudes did not correlate with the probability of disclosing certain type of information on OSNs. Perhaps exploration of the PDA constructs may assist researchers in examining the relationships

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between individuals' personal data concerns and their claimed personal interest and protective behaviours. For example, 91 per cent of the participants in the study conducted by Sato (2010) indicated the need for a system that will enable them to control how their data are used. Meanwhile, researchers report fewer people actually adopting existing security and privacy mechanisms to protect their personal data online (EU Commission, 2011, Shelton *et al.*, 2015). The research findings by Shelton *et al.* (2015) highlight usability and lack of awareness of existing security mechanisms as the two main factors hindering people's ability to be more actively involved in the protection of their personal data and privacy. The usable security and privacy research community could therefore identify specific modifications of PDAs, behaviours and skills that can foster a more positive appreciation for personal data through an in-depth exploration of the constructs identified.

The findings mostly reflect the underlying themes explored through studies focused on personal data, rather than those focused on privacy concerns. Thus, whereas privacy measurement scales tend to focus on constructs such as collection, control, errors, authorized use and awareness of privacy practices, the personal data literature is more concerned with issues relating to security, availability, privacy, risk and benefits. For instance, Kobsa (2007) suggest that although consumers appreciate the benefits of user profiling and personalization, they are not willing to be profiled due to privacy concerns. Sato (2010), on the other hand, surveyed about 3,000 people from six different countries and concluded that even though people are generally concerned about the privacy of their data, they are more positive about the potential benefits which they would normally weigh against the risk. Accordingly, when the benefits outweigh the perceived risk, cloud services users become more open to the data sharing concept. Although most people now accept that life in the digital age involves disclosure of personal data, concerns remain about the actual use of the data (EU Commission, 2011). However, as Acquisti *et al.* (2016) pointed out, because the experience individuals may have when their personal information is exposed may differ, their concerns about the use of their personal data also tend to vary. A PDA scale will therefore be required to capture these individual differences to enable more representative user-models for the design of cybersecurity tools.

The findings of the study have implications for both research and design practices in the field of usable security and privacy. Because little prior research exists specifically on scale development for PDA, this study signifies the first empirical examination of the concept. Existing instruments attempting to assess attitudes towards personal data are mainly based on privacy-focused design which have produced a variety of attitudes ranging from one to six dimensions. Although there are significant similarities in the attitudes between privacy and personal data, privacy instruments tend not to relate specifically to attitudes towards what people may view as personal information. Information privacy, one of the most vital aspects of privacy, is concerned with protecting personal data of individuals. However, the range of potential implications in relation to the collection and sharing of personal data goes beyond the issue of privacy and includes constructs related to responsibilities, security, risk and benefits as explored in this study. All stakeholders within the digital economy, need to carefully consider these dynamics to ensure that they understand and are willing to accept risk reward balance of PDE (Kalapesi, 2013). Consequently, the study makes a major contribution to the growing body of literature on user modelling in the field of information security by highlighting the potential of including PDA as a determinant in predictive user models necessary for the design of adaptive cybersecurity.

In summary, the findings are believed to provide two major contributions to information security research and design practices:

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(1) a framework describing the primary dimensions of individuals' attitude towards personal data; and
(2) an instrument that can easily be modified and used to measure those concerns and preferences for adaptive security designs.

As indicated earlier, there is relatively little research literature on attitudes towards personal data that deal specifically with how individuals view the construct of personal data. There is therefore relatively little information available to guide designers in addressing personal data issues when designing new interfaces and technologies. A lot of new technologies and services have implications for how personal information is handled and how people react to them. Therefore, the existence of an instrument such as the PDA scale has the potential for distinctively detecting attitude profiles of technology users that can be very useful for adaptive cybersecurity designs. Essentially, the six factors identified provide a framework around which personal data discussions and models might be developed by HCI researchers and information system designers. Thus, human factor engineers and designers may find such a tool very useful in looking to personalized interfaces where personal data issues are pertinent.

5.1 Study limitations and future directions

Although the findings presented in this report form an effective first draft of a personal data attitude instrument, several limitations of the study need to be highlighted. Notably, the convenience and accidental sampling methods adopted may limit the external validity of the findings. These preliminary data have however been used to provide empirical evidence in support of the PDA scale's potential to be a valuable cybersecurity research and design tool. However, further research work need to be carried out with different types of populations, to establish the external validity of the PDA instruments. We recognize that our clustering approach is relatively basic and plan to extend the data collection towards refining the clusters as part of our future work. Nevertheless, the primary contribution of this work is to demonstrate the feasibility of segmenting users based on their attitude towards personal data among other determinants.

The evaluation presented in this paper is the first step towards developing a robust empirical evidence of the PDA dimensions. Future research with more broader samples are required to replicate the factor structure and validate inferences that can be made based on the PDA scores. Variables like, technology users' personal data attitude change overtime and the underlying factors such as local context and/or cultural differences, could then be examined within a single integrated analysis using machine learning techniques and structural equation modelling. Finally, the findings of this study could be incorporated into the design of user models required to personalize cybersecurity or privacy education and products to different category of internet users.

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